

KENYA NATIONAL EXAMINATIONS COUNCIL
GRADE 7 SCHOOL-BASED ASSESSMENT (SBA) 2026
COMPREHENSIVE SOLVED QUESTION & ANSWER MANUAL

Note: This document provides complete, step-by-step worked solutions for all Grade 7 SBA examination papers across eight core learning areas: Mathematics, English, Kiswahili, Christian Religious Education (CRE), Integrated Science, Pre-Technical Studies, Creative Arts & Sports (CAS), and Social Studies (SST). Islamic Religious Education (IRE) has been excluded as requested.

1. MATHEMATICS (GRADE 7 SBA 2026)

Q1 (a): A non-governmental organization donated 382 459 000 shillings to the ministry of health for public hospital renovations. Write the amount in words.

Answer: Three hundred eighty-two million, four hundred fifty-nine thousand shillings.

Q1 (b): Determine the total value of digit 4 in 382 459 000.

Answer: 400,000 (Four hundred thousand).

Q2: Medals were packed in 6 boxes each containing 8 medals. During awarding, 10 medals were found broken. Determine total medals presented.

Answer: Total medals packed = $6 \times 8 = 48$ medals. Medals presented = $48 - 10 = 38$ medals.

Q3 (a): A farmer harvested 48 mangoes and 60 oranges and packed them into identical baskets with no fruit left over. Find the highest number of baskets obtained.

Answer: Greatest Common Divisor (GCD) of 48 and 60:

Factors of 48 = $2^4 \times 3$

Factors of 60 = $2^2 \times 3 \times 5$

$\text{GCD}(48, 60) = 2^2 \times 3 = 12$ baskets.

Q3 (b): Determine the number of mangoes in each basket.

Answer: $48 \div 12 = 4$ mangoes per basket.

Q4: Omogeni packs juice in plastic containers holding $\frac{1}{4}$ litres each. He sold 24 litres. How many containers did he sell?

Answer: $24 \div (\frac{1}{4}) = 24 \times 4 = 96$ plastic containers.

Q5: A tailor bought 34.8 m of fabric. Each costume requires 2.32 m. Calculate total costumes made.

Answer: $34.8 \div 2.32 = 15$ costumes.

Q6: A square sitting room floor carpet has an area of 40.96 m². Determine the length of one side using division method.

Answer: Length = $\sqrt{40.96} = 6.4$ meters.

Q7: Learners ran around a pitch of length (3x + 2) m and width (2x + 4) m. Write a simplified expression for the total perimeter.

Answer: Perimeter = $2 \times (\text{Length} + \text{Width}) = 2 \times [(3x + 2) + (2x + 4)] = 2 \times (5x + 6) = (10x + 12)$ meters.

Q8: A father is 18 years older than his son. The sum of their present ages is 46 years. Determine the son's age.

Answer: Let son's age be y. Father's age = y + 18. Equation: $y + (y + 18) = 46 \Rightarrow 2y = 28 \Rightarrow y = 14$ years.

Q9: Write two inequality statements for x learners going on a field trip where the number must be at least 20 but not more than 45.

Answer: 1) $x \geq 20$

2) $x \leq 45$ (Combined: $20 \leq x \leq 45$).

Q10: A granary ramp reaches a height of 4 m and has a length of 10 m. Calculate the horizontal base length x.

Answer: By Pythagoras' theorem: $x^2 + 4^2 = 10^2 \Rightarrow x^2 + 16 = 100 \Rightarrow x^2 = 84 \Rightarrow x = \sqrt{84} \approx 9.17$ meters.

Q11: Determine the perimeter of a birthday hat marker comprising a semicircle of diameter 7 cm and two straight sides of 7.5 cm.

Answer: Arc length = $\frac{1}{2} \times \pi \times d = \frac{1}{2} \times (22/7) \times 7 = 11$ cm. Total Perimeter = $11 + 7.5 + 7.5 = 26$ cm.

Q12: A vegetable garden consists of a rectangle (42 m × 28 m) bounded by two semicircles of diameter 28 m. Calculate total area.

Answer: Rectangular Area = $42 \times 28 = 1176$ m². Two semicircles = 1 full circle of radius 14 m = $(22/7) \times 14 \times 14 = 616$ m². Total Area = $1176 + 616 = 1,792$ m².

Q13: A cylindrical water tank has diameter 7 m, height 10 m, and is 3/4 full. Determine water volume in litres.

Answer: Total Volume = $\pi \times r^2 \times h = (22/7) \times 3.5 \times 3.5 \times 10 = 385$ m³. Water Volume = $3/4 \times 385 = 288.75$ m³ = 288,750 litres.

Q14: An odometer read 90 176 km initially and 90 296 km at the end. The trip took 1 hr 30 min. Calculate speed in km/h.

Answer: Distance = $90\,296 - 90\,176 = 120$ km. Time = 1.5 hours. Speed = $120 \div 1.5 = 80$ km/h.

Q15: Convert a recorded tea temperature of 336 Kelvin into degrees Celsius.

Answer: Temperature in °C = $336 - 273 = 63^{\circ}\text{C}$.

Q16: Mwende bought a bag of beans at Ksh. 9 000 and sold it at a 20% profit. Calculate the selling price.

Answer: Profit = $20\% \times 9000 = \text{Ksh. } 1,800$. Selling Price = $9000 + 1800 = \text{Ksh. } 10,800$.

Q17: A sales agent earned Ksh. 8 000 commission on sales worth Ksh. 40 000. Determine the commission percentage.

Answer: Commission % = $(8000 \div 40000) \times 100\% = 20\%$.

Q18: A triangular table top has internal angles $(n - 30)^{\circ}$, n° , and $(2n + 10)^{\circ}$. Determine the difference between the largest and smallest angles.

Answer: Sum = $180^{\circ} \Rightarrow (n - 30) + n + (2n + 10) = 180 \Rightarrow 4n - 20 = 180 \Rightarrow n = 50^{\circ}$. Angles: Smallest = 20° , Middle = 50° , Largest = 110° . Difference = $110^{\circ} - 20^{\circ} = 90^{\circ}$.

Q19: Construct triangle ABC with AB = 8 cm, $\angle ABC = 45^{\circ}$, $\angle CAB = 30^{\circ}$. Determine length of BC.

Answer: Using Sine Rule: $BC / \sin(30^{\circ}) = 8 / \sin(105^{\circ})$. Length BC ≈ 4.14 cm.

Q20: Frequency Distribution Table for Learner Masses:

Mass (kg)	Tally	Frequency
30		1
35		3
40		6
45		2
50		2

2. ENGLISH (GRADE 7 SBA 2026)

QTask 1 Part I: Q1: What will you say when you welcome a new learner?

Answer: Welcome to our school! We are very happy to have you join our Grade 7 class.

QTask 1 Part I: Q2: What will you tell the learner about your school?

Answer: I will tell the learner about our daily school routines, core subjects, supportive teachers, and co-curricular activities.

QTask 1 Part I: Q3: Which place in your school would you show the learner first? Why?

Answer: Our Grade 7 classroom, so that they know where our class sits and can feel settled right away.

QTask 1 Part I: Q4: Why is it important to make new learners feel welcome?

Answer: It helps reduce anxiety, builds a sense of belonging, and allows them to adjust quickly to learning.

QTask 1 Part II: Q5: Who called Mutanga in the morning?

Answer: His grandmother.

QTask 1 Part II: Q6: What did Mutanga take to his sister?

Answer: A cup of tea.

QTask 1 Part II: Q7: Why did Mutanga walk quietly to the bedroom?

Answer: Because his younger sister was sleeping and he did not want to disturb her.

QTask 3: Q1 (a): When did Asha and her brother arrive at the farm?

Answer: Early in the morning.

QTask 3: Q1 (b): Why do you think Asha counted the eggs carefully?

Answer: To ensure none were broken or missing and to keep an accurate count for her grandmother.

QTask 3: Q1 (c): Why did the children help their grandmother carry vegetables?

Answer: To assist her with farm chores and lighten her workload.

QTask 3: Q1 (d): From the story, the children are _____.

Answer: Helpful.

QTask 3: Q2: In about 60 words, summarise the benefits of reading for pleasure.

Answer: Reading for pleasure expands vocabulary and improves language usage. It stimulates imagination and exposes learners to diverse ideas, places, and cultures. Regular reading enhances spelling and writing skills by presenting correct sentence structures. Additionally, reading storybooks and digital texts makes learning enjoyable and fosters lifelong critical thinking habits.

QTask 4: Q3 (a-c): Fill in blanks with correct form of words in brackets: (patient, confident, excite)

Answer: a) patiently

b) confidently

c) excitement

QTask 4: Q4 (a-b): Replace underlined words with correct form of words in brackets: (put off, give out)

Answer: a) The committee decided to put off the meeting.
b) Please give out the exercise books to the class.

QTask 4: Q5 (a-c): Fill in blanks with correct alternative: (was/were, has/have, is/are)

Answer: a) were
b) has
c) is

QTask 4: Q6 (a-d): Fill in blanks with correct adjective form: (cheer, danger, old, order)

Answer: a) cheerful
b) dangerous
c) oldest
d) orderly

QTask 4: Q7 (a-c): Fill in blanks with correct prepositions: (on/in/at, among/between)

Answer: a) on
b) among
c) between

QTask 5: Q8 (a-d): Lion's Animal Meeting Story analysis questions.

Answer: a) Lion called the meeting to discuss protecting the water source during the dry season.
b) The animals held a formal meeting and proposed constructive solutions.
c) A happy, smiling, and approving facial expression.
d) Unity and cooperation enable communities to overcome shared challenges.

QTask 5: Q9 (a-c): Poem analysis: 'MY SCHOOL'

Answer: a) 2 stanzas.
b) Rhyming pairs: trees - breeze, play - day (or grow - know, bright - delight).
c) Learners respect / guide by their teachers.

QTask 5: Q10 (a-e): Cultural Visit Story analysis questions.

Answer: a) Setting: A cultural centre.
b) Style feature: Dialogue.
c) Musa and Neema are curious / attentive / polite.
d) Theme of education is shown through examining historical photos and asking questions.
e) Value shown at the end: Gratitude (thanking their teacher).

QTask 6: Q11: Composition writing prompt: 'The task seemed difficult until we decided to work together.'

Answer: Model Outline: Begin by describing a daunting assigned project (e.g., clearing the overgrown school environmental garden). Describe how individual efforts felt exhausting and ineffective. Detail how learners organized into teams, allocated duties according to strengths, and encouraged one another. End with the rapid, successful completion of the project and a reflection on the power of synergy and teamwork.

3. KISWAHILI (GREDI YA 7 SBA 2026)

QZoezi 1A: Swali 1: Taja jambo moja unaloweza kufanya vizuri shuleni?

Answer: Kusoma kwa sauti, kuchora picha, kufanya hesabu, au kucheza mpira wa miguu.

QZoezi 1A: Swali 2: Mwalimu amekusifu mbele ya darasa. Utahisi vipi na kwa nini?

Answer: Nitahisi furaha na kujiamini kwa sababu juhudi zangu zimetambuliwa.

QZoezi 1A: Swali 3: Ukishindwa kufanya vizuri katika shughuli fulani darasani, utafanya nini?

Answer: Nitaomba msaada kutoka kwa mwalimu au wenzangu na kufanya mazoezi zaidi.

QZoezi 1A: Swali 4: Utampa ushauri gani rafiki asiyejiamini masomoni?

Answer: Nitamshauri kuwa kila mtu ana uwezo wa kipekee, na akijiamini na kutia bidii atafanikiwa.

QZoezi 1B: Swali 5: Jamii hutumia vifaa vya kidijitali kwa manufaa gani?

Answer: Kuwasiliana kwa haraka, kujifunza mtandaoni, na kuwasiliana na familia na marafiki wa mbali.

QZoezi 1B: Swali 6: Mwalimu anaweza kutumia vipi vifaa vya kidijitali kufundishia?

Answer: Kutoa maelekezo, kuonyesha vielelezo vya masomo, na kuwafikia wanafunzi kwa urahisi mtandaoni.

QZoezi 1B: Swali 7: Nini kitatokea ukitoa taarifa za binafsi mtandaoni?

Answer: Faragha yangu itaingiliwa, na ninaweza kutapeliwa au taarifa zangu kutumiwa vibaya na wahalifu.

QZoezi 1B: Swali 8: Neno jingine lenye maana sawa na makini ni:

Answer: Uangalifu / utulivu.

QZoezi 3: Swali 1: Majukumu ya watoto hutegemea nini kulingana na aya ya kwanza?

Answer: Umri na uwezo wa watoto.

QZoezi 3: Swali 2: Nini kitatokea mazingira hayatatunzwa?

Answer: Magonjwa yatasambaa na afya ya viumbe hai itaathirika vibaya.

QZoezi 3: Swali 3: Watoto wanachangia vipi katika mshikamano wa jamii?

Answer: Kwa kuwa na ushirikiano mzuri unaowawezesha kujenga urafiki na kuepuka vitendo viovu.

QZoezi 3: Swali 4: Maana ya neno 'kutekeleza' kulingana na kifungu:

Answer: Kutimiza, kufanya, au kutenda wajibu.

QZoezi 3: Swali 5-8: Maswali ya Usalama Shuleni (1. Kufuata sheria, 2. Wajibu wa utawala, 3. Mazingira machafu, 4. Kinga ni bora kuliko tiba)

Answer: 5. Kufuata sheria za shule, kuepuka michezo hatari, na kuripoti utovu wa nidhamu.

6. Kuweka uzio na lango imara na kuweka vifaa vya kuzimia moto.

7. Ajali zitatokea (kama kuteleza) na magonjwa yatasambaa.

8. Kuchukua hatua za kujikinga na hatari kabla haijatokea.

QZoezi 4: Swali 9-12: Ufupisho wa Mila Potofu na Athari zake

Answer: 9. Kubaguliwa, kunyimwa haki, na kuoleswa mchanga.

10. Kuacha shule, kuathirika kisaikolojia, kiafya, na kutoroka.

11. Elimu dhidi ya mila potofu inaleta usawa, elimu, na maendeleo jamiani.

12. Maana ya 'dhiki chungu nzima': Shida au mateso mengi sana.

QZoezi 5: Swali 13-17: Chaguzi za Sarufi (13. alitufundisha, 14. kutolea, 15. inayotumia, 16. ya, 17. :)

Answer: 13. B. alitufundisha

14. D. kutolea

15. A. inayotumia

16. A. ya

17. B. . (au :)

QZoezi 5: Swali 18-27: Mazoezi ya Sarufi na Lugha

Answer: 18. Vumbi ni nomino ya kuhesabika/jamii.

19. Wakati uliopo: Kananu anapalilia maua vizuri.

20. Kinyume cha mfalme: malkia.

21. Kuanza na Babu: Babu alipikiwa chai na Nuno.

22. Ukanusho: Wanafunzi watakuwa hawasafishi darasa kesho asubuhi.

23. Ngeli: Jiko (LI-YA), Chumba (KI-VI).

24. Kitenzi kishirikishi: ndiye.

25. Sentensi sahili: Baba anasoma gazeti.

26. Ukubwa wa kikombe: Kombe hili kubwa lina chai nyingi.

27. Usemi wa taarifa: Kiranja alisema kuwa wangepanda miche ya matunda siku hiyo jioni.

QZoezi 6: Swali 28: Barua ya kirafiki ya kumwalika rafiki katika sherehe ya kutoa shukrani.

Answer: Muundo wa Barua: Anwani ya mwandishi, tarehe, mtajo (Kwa Rafiki Mpendwa...), maelezo kuhusu ushindi wa tuzo la mwigizaji bora, mwaliko na maelezo ya sherehe, na hitimisho rasmi la kirafiki.

QZoezi 7: Swali 29: Uchambuzi wa Wimbo wa Kazi (Shikeni majembe...)

Answer: a) Mazingira ya kilimo / shambani (kutaja majembe, magugu, mbolea).

b) Ujumbe: Kutia mbolea ili mimea inawiri na kuleta mavuno mengi ya kuondoa njaa.

c) Mbinu za kuwasilisha: Kutumia ala za muziki, ishara za mwili, na kuimba kwa mdundo wa pamoja.

d) Sifa za jamii: Ni jamii yenye bidii na inayotegemea kilimo.

QZoezi 7: Swali 30: Uchambuzi wa Novela (Mtemi Makini na Jamii ya Wapambe)

Answer: a) Uongozi mwema: Mtemi Makini alitafuta njia mbadala (madini) uchumi ulipodorora na aliandaa vikao vya hamasa badala ya kutumia nguvu.

b) Matukio: Wakazi kukataa kuhamishwa, Mtemi Makini kuandaa vikao vya hamasa, na kujengwa kwa barabara, shule na hospitali.

c) Mandhari: Mandhari ya mahali (Machangani/Baraka) na mandhari ya wakati.

d) Mbinu za lugha: (i) Istiari: 'bidii yao ya mchwa', (ii) Tasbihi: 'kama safu ya askari', (iii) Methali: 'Hayawi hayawi huwa'.

4. CHRISTIAN RELIGIOUS EDUCATION - CRE (GRADE 7 SBA 2026)

Q1: State two differences between the two accounts of creation in Genesis 1 and 2.

Answer: 1) In Genesis 1, creation occurs in 6 structured days, whereas in Genesis 2 no duration is mentioned.

2) In Genesis 1, man and woman were created simultaneously last on the 6th day, whereas in Genesis 2 man was created first, followed by plants/animals, and woman was created from man's rib later.

Q2: Identify two responsibilities God gave to human beings in Genesis 1 and 2.

Answer: 1) To cultivate, care for, and guard the Garden of Eden and the environment.

2) To name all created animals and exercise dominion over all living creatures.

Q3: Outline two ways children can care for domestic birds at home.

Answer: 1) Providing them with clean food and fresh water regularly.

2) Constructing clean, safe shelters and protecting them from predators.

Q4: Write two ways in which environmental conservation is important to people in Kenya today.

Answer: 1) Protects water catchment areas and preserves soil fertility for agriculture.
2) Reduces pollution and mitigates climate change risks like droughts and floods.

Q5: Identify three occasions during which the Bible is read by Christians in Kenya.

Answer: 1) Sunday church worship services.
2) Weddings and baptismal ceremonies.
3) Funeral services and daily family devotions.

Q6: Select a box that contains only books of the New Testament (Box E: Ezra, Job, James | Box F: Ruth, Esther, Nehemiah | Box G: Timothy, Titus, Philemon).

Answer: Box G (Timothy, Titus, Philemon).

Q7: Write two ways in which God prepared Moses before sending him to deliver the Israelites from Egypt.

Answer: 1) Raised in Pharaoh's palace where he acquired leadership and Egyptian administrative skills.
2) Tended Jethro's flock in the Midian wilderness, developing patience and familiarity with the desert route.

Q8: State two values that learners can acquire from the teachings of John the Baptist.

Answer: 1) Honesty / Fairness (advising tax collectors and soldiers to be content and just).
2) Humility (pointing people toward Jesus Christ).

Q9: Outline two qualities the angel Gabriel revealed about Jesus during the annunciation to Mary.

Answer: 1) He will be great and will be called the Son of the Most High.
2) His kingdom will last forever (Eternal King).

Q10: Write two reasons why it is necessary to pray regularly.

Answer: 1) To maintain a close personal relationship with God and seek His guidance.
2) To give thanks for blessings and seek forgiveness and divine protection.

Q11: State two points on the role of the church in promoting human health in Kenya today.

Answer: 1) Establishing and operating mission hospitals, dispensaries, and clinics.
2) Conducting community health education, counseling, and free medical outreach camps.

Q12: Identify two consequences of learners engaging in sex before marriage.

Answer: 1) Unintended teenage pregnancies leading to school dropouts.
2) Increased risk of contracting Sexually Transmitted Infections (STIs) and HIV/AIDS.

Q13: State two Christian teachings on marriage.

Answer: 1) Marriage is a sacred union instituted by God between a man and a woman.
2) Marriage is permanent, monogamous, and based on mutual love and respect.

Q14: Give two reasons why some young people in Kenya abuse drugs today.

Answer: 1) Peer pressure and the desire for social acceptance.
2) Stress, depression, lack of employment, or curiosity.

Q15: Identify two negative effects of gambling activities on learners.

Answer: 1) Financial loss leading to debt or temptations to steal.
2) Addiction resulting in poor academic performance, truancy, and loss of focus.

Q16: State four benefits of learning CRE in school.

Answer: 1) Promotes moral uprightness and character development.
2) Nurtures spiritual growth and a personal relationship with God.
3) Fosters national unity, tolerance, and peaceful coexistence.
4) Helps learners make ethical decisions when facing life challenges.

Q17: Outline four effects of the translation of the Bible into local languages in Kenya.

Answer: 1) Made the Word of God easily accessible and understandable to local communities.
2) Accelerated the growth of local churches and indigenous Christian leadership.
3) Promoted literacy and preservation of local African languages.
4) Enabled deeper contextual understanding and application of Scripture.

Q18: Give four reasons why Christian parents present their children for dedication in Church.

Answer: 1) To express gratitude to God for the gift of a child.
2) To commit the child's life to God's protection and blessing.
3) To publicly pledge to raise the child according to Christian principles.
4) To seek prayers and spiritual support from the church community.

Q19: Identify four worship practices performed both at home and in church.

Answer: 1) Reading and studying Holy Scriptures.
2) Singing praise and worship songs and hymns.
3) Offering individual and intercessory prayers.
4) Giving tithes and offerings.

Q20: Give four ways in which people misuse social media platforms in Kenya today.

Answer: 1) Spreading fake news, unverified rumors, and hate speech.
2) Engaging in cyberbullying, trolling, and online harassment.
3) Sharing explicit, pornographic, or immoral content.
4) Executing financial scams and wasting constructive learning time.

5. INTEGRATED SCIENCE (INTSCI GRADE 7 SBA 2026)

Q1 (a)(i): State the factor affecting plant growth that was changed in this investigation.

Answer: Amount of sunlight (Sunlight intensity).

Q1 (a)(ii): Explain your answer in (i) above.

Answer: Plant A was placed in full sunlight, Plant B in moderate sunlight, and Plant C in low sunlight.

Q1 (b)(i): Identify two factors that the learners did not change.

Answer: 1) Type and amount of soil in the pots.

2) Volume and frequency of water provided.

Q1 (b)(ii): Explain why keeping these factors constant was important.

Answer: To ensure a fair test, so that any observed difference in plant growth is strictly caused by the variation in sunlight.

Q1 (c)(i): State the conclusion the learners made from these results (Plant A: 18 cm, Plant B: 12 cm, Plant C: 5 cm).

Answer: Increased sunlight promotes faster and healthier growth in bean plants.

Q1 (c)(ii): Give a reason for your answer in (i).

Answer: Plant A receiving full sunlight grew the tallest (18 cm), whereas Plant C in low sunlight grew the least (5 cm).

Q1 (d): A learner suggests watering Plant C more than others to help it grow faster. Explain how this affects the investigation.

Answer: It ruins the controlled nature of the experiment by introducing a second independent variable (water volume), making it impossible to attribute growth differences solely to sunlight.

Q2 (a): A container has layers of sand, salt, iron filings, and plastic beads. Is adding water the best first step?

Answer: No. Adding water first causes iron filings to rust and makes sand and iron filings difficult to separate. A magnet should be used first while the mixture is dry to easily remove all iron filings.

Q2 (b): Describe the complete steps to separate and recover all four substances.

Answer: 1) Magnetic Separation: Pass a magnet over the dry mixture to attract and recover iron filings.

2) Flotation: Add water to the remaining mixture. Plastic beads float to the top; skim them off.

3) Filtration: Pour the remaining mixture through filter paper to trap insoluble sand as residue.

4) Evaporation: Heat the salt-water filtrate to evaporate water, leaving pure salt crystals behind.

Q3 (a): Solution R leaves blue litmus blue and red litmus red. A learner claims R must be a base. Is the learner right?

Answer: No, the learner is incorrect. Solution R is neutral (such as pure water), because neutral solutions do not change the color of either red or blue litmus paper.

Q3 (b)(i-ii): The learner mixes equal amounts of solution P (acid) and Q (base). Predict and explain how acidity changes.

Answer: i) The acidity of the mixture decreases towards neutral (pH 7).

ii) Explanation: P is acidic (turns blue litmus red) and Q is basic (turns red litmus blue). When mixed in equal amounts, neutralization occurs.

Q3 (c): Why is tasting P, Q, and R unsafe? Suggest a safer scientific method.

Answer: Tasting is unsafe because unknown lab solutions may be toxic, corrosive, or poisonous. A safer scientific method is using a universal indicator or a calibrated digital pH meter.

Q3 (d): Explain why litmus paper alone may not always provide enough information about acid/base strength.

Answer: Litmus paper only provides a qualitative result (acidic or basic) but does not indicate the quantitative pH value or concentration strength of the solution.

Q4 (a-c): Adolescence and Reproductive System analysis.

Answer: a) Evidence: Start of menstruation in girls indicates ovulation and hormone production, confirming active reproductive system function.

b) Menstruation occurs only in girls because females possess a uterus and ovaries designed for gestation; menstruation is the monthly shedding of the unfertilized uterine lining.

c) Active Organ Role: Ovaries produce mature ova and hormones (estrogen/progesterone) OR Testes produce sperm cells and testosterone.

Q5 (a-d): Kidney function and body water balance during physical exertion.

Answer: a) Kidney Role: Regulates osmoregulation (blood water and salt balance) and excretes nitrogenous waste.

b) Explanation: In dehydration, the brain triggers ADH release, causing kidneys to reabsorb maximum water back into blood, yielding concentrated, low-volume, dark urine.

c) i) Disagree with 'more waste'. ii) Reason: The dark color is caused by high concentration (low water content), not an increased quantity of waste.

d) Other response: Reduced sweat production and triggering the thirst sensation to encourage drinking.

Q6 (a-c): Circuit analysis with cell, switch, and bulb.

Answer: a) Learner's claim ('switch produces electricity') is incorrect. The cell is the source of electrical energy; the switch only completes or opens the circuit.

b) If wire is cut in Circuit B, the bulb turns off immediately because the circuit is broken, stopping current flow.

c) Repairing cut wire while cell is connected is unsafe because working on live circuits risks electric shock or short circuits.

Q7 (a-c): Magnet P and Q behavior.

Answer: a) Non-contact force: Like magnetic poles (N-N) exert a magnetic field force of repulsion even without physical contact.

b) To make them move together: Reverse one magnet so opposite poles face each other (North facing South); opposite poles attract.

c) Stronger repulsion: Bring the same pole of a third, stronger magnet near the repelling poles or stack it behind one magnet to boost field strength.

6. PRE-TECHNICAL STUDIES (PRETECHNICAL GRADE 7 SBA 2026)

QTask 1: Q1: Explain three reasons for studying Pre-Technical Studies in Junior School.

Answer: 1) Equips learners with foundational practical and technical skills for everyday problem-solving.

2) Exposes learners to diverse technical and vocational career options early in education.

3) Instills safety awareness, tool maintenance culture, and creative engineering thinking.

QTask 1: Q2: Identify three technical careers learners can pursue.

Answer: 1) Electrical / Mechanical / Civil Engineer or Technician.

2) Architectural Drafter / Technical Designer.

3) Computer Hardware Specialist / Network Technician.

QTask 1: Q3: Identify lines P, Q, R, S on technical drawing and explain what each represents.

Answer: P (Center Line): Alternate long and short dashes; represents axes of symmetry or centers of circular features.

Q (Outline / Continuous Thick Line): Represents visible edges and contours of an object.

R (Hidden Detail Line): Short dashed line; represents concealed internal edges or surfaces.

S (Dimension Line / Extension Line): Continuous thin line with arrowheads; indicates exact measurements.

QTask 1: Q4: Identify materials used to make items: Geometrical set, Ruler, Desk waste, Broken windowpanes.

Answer: 1) Geometrical set: Metal / Plastic
2) Ruler: Plastic / Stainless Steel / Wood
3) Desk waste parts: Timber / Wood
4) Broken windowpanes: Glass

QTask 1: Q5: State three ways of storing tools safely in a workshop.

Answer: 1) Clean tools and return them to designated racks, shadow boards, or toolboxes after use.
2) Inspect tools regularly and never store damaged or dull cutting tools.
3) Apply a thin coat of rust-preventive oil on metal working surfaces before storage.

QTask 1: Q6: Write drawing abbreviations in full: R, NTS, CL.

Answer: R = Radius
NTS = Not To Scale
CL = Center Line

QTask 1: Q7: State the use of tools: Scriber, Measuring tape, Handsaw, Hand file.

Answer: 1) Scriber: Used for marking precise layout lines on metal sheet surfaces.
2) Measuring tape: Used for measuring linear dimensions over long distances.
3) Handsaw: Used for cutting wooden materials and timber.
4) Hand file: Used for smoothing, deburring, and shaping metal or wood edges.

QTask 1: Q8: State three properties of metals used to distinguish them from non-metals.

Answer: 1) High electrical and thermal conductivity.
2) High density, strength, and metallic luster (shininess).
3) Malleability (can be hammered into sheets) and ductility (can be drawn into wires).

QTask 2: Q9: Explain three factors to consider in determining the size of production units.

Answer: 1) Amount of capital invested in the business.
2) Level of technology, equipment, and machinery utilized.
3) Scale of output capacity and size of the market demand.

QTask 2: Q10: Describe three entrepreneurial qualities demonstrated by Mwanahamisi (bakery owner).

Answer: 1) Innovation & Creativity: Introducing mini-cakes and doughnuts when business is slow.
2) Perseverance & Resilience: Working long hours and persisting despite rising ingredient

prices.

3) Good Customer Care: Treating customers politely and maintaining quality service.

QTask 2: Q11: Set three financial goals for personal development (Amani's savings).

Answer: 1) Short-term goal: Save a fixed portion of weekly allowance to buy a new school bag.

2) Medium-term goal: Build an emergency fund for school stationery and project supplies.

3) Long-term goal: Develop disciplined budgeting to avoid impulse spending on snacks.

QTask 3: Q12 (a-b): Cyber security threat on public Wi-Fi (Jabali).

Answer: a) Threat: Data interception / Man-in-the-Middle (MitM) attack / Malware infection over unsecured public network.

b) Protection: Use a Virtual Private Network (VPN), enable personal firewall, and avoid sensitive transactions on public Wi-Fi.

QTask 3: Q13 (a-b): Select ICT communication tools for awareness campaign and explain benefits.

Answer: a) Tools: Public Address Systems / Loudspeakers, Community WhatsApp groups / Social Media, and Bulk SMS services.

b) Benefits: Enables rapid, widespread reaching of residents at minimal cost and allows immediate two-way community feedback.

QTask 3: Q14 (a-d): Identify appropriate hardware: Data transfer, Hard copy reports, Performance capture, Score display.

Answer: a) Data transfer: USB Flash Drive / External Hard Drive.

b) Hard copy reports: Printer.

c) Performance capture: Digital Video Camera / Camcorder.

d) Score display: LED Digital Scoreboard / Large Display Monitor.

QTask 3: Q15 (a-b): Suitable computer for April's fieldwork travel.

Answer: a) Suitable computer: Laptop / Notebook / Tablet computer.

b) Reasons: 1) Portable and lightweight for easy transport during field movement. 2) Built-in rechargeable battery and integrated wireless connectivity for typing and transmitting daily reports.

7. CREATIVE ARTS AND SPORTS - CAS (GRADE 7 SBA 2026)

QPart I: Q1: The rhythm is in 2/4 time. How are bar lines inserted?

Answer: Bar lines are placed after every 2 crotchet beats (or equivalent quaver/semiquaver groupings) to divide the rhythm into complete 2-beat measures.

QPart I: Q2: Write the letter names of the four treble clef notes.

Answer: (i) F

(ii) D

(iii) C

(iv) G

QPart I: Q3: Name the musical note that corresponds to each rest symbol shown.

Answer: (i) Semibreve rest (Whole rest)

(ii) Minim rest (Half rest)

(iii) Crotchet rest (Quarter rest)

(iv) Quaver rest (Eighth rest)

QPart I: Q4: Compose a four-bar melody in C Major in 2/4 time using crotchets and quavers on staff.

Answer: Structure: Bar 1 (C C) | Bar 2 (D E) | Bar 3 (F D) | Bar 4 (C - minim/crotchet rest). Begins and resolves on tonic C.

QPart I: Q5: Create a short 6-sentence story about caring for the environment.

Answer: Beginning: Juma noticed piles of discarded plastic bottles clogging the school drainage channel. Middle: He rallied his classmates to clear the blockage and collect the recyclable plastic. Solution: They established a school green club and planted colorful flower beds along the cleared path. Ending: The school compound became clean, vibrant, and flood-free. Character: Juma. Storytelling technique: Using varied vocal pitch and animated gestures during presentation.

QPart II: Q6 (a-b): Physical fitness component identification from diagram.

Answer: a) Component: Flexibility / Core Strength / Balance.

b) Importance: Enhances range of joint motion, improves movement efficiency, and reduces risk of muscle strains during athletic performance.

QPart II: Q7 (a-b): Javelin throwing technique and phases.

Answer: a) Correct landing action: Release the javelin at a 30°-35° elevation angle with proper apex trajectory so the tip strikes the ground first.

b) Three phases in order: 1) Approach run (Carry and acceleration phase) 2) 5-stride crossover transition phase 3) Delivery and recovery phase.

QPart II: Q8 (a-b): Handball passing and possession tactics.

Answer: a) Pass type: Bounce pass or Jump pass to bypass closely marking defenders.

b) Retaining possession: Fast short passes, rapid positional dodging/feinting, and creating open space.

QPart II: Q9 (a-b): Football ball-trapping technique above waist level.

Answer: a) Technique: Chest trapping.

b) Execution: Move into line with the incoming ball, arch the torso back slightly to absorb impact with chest muscles, and cushion the ball down to the feet.

QPart II: Q10: State two ways players demonstrate fairness during games.

Answer: 1) Respecting referee calls promptly without arguments.

2) Showing sportsmanship by assisting fallen opponents and shaking hands after the match.

QPart II: Q11 (Option 2): Tug-of-war tactics against Grade 8 team.

Answer: 1) Maintain a low center of gravity with feet wide apart and heels planted firmly in the soil.

2) Pull together using synchronized, powerful backward leg drives on rhythmic team counts.

QPart III: Q12: Explain two reasons why warm colors are suitable for a cultural festival poster.

Answer: 1) Warm colors (red, orange, yellow) capture immediate visual attention from a distance.

2) They convey energy, excitement, warmth, and festive celebration appropriate for cultural events.

QPart III: Q13 (a-b): Type of balance shown in visual design picture.

Answer: a) Type of balance: Asymmetrical (Informal) balance.

b) Reason: Visual elements of differing shapes and colors are distributed unevenly across axes while maintaining equal visual weight.

QPart III: Q14: Suggest two ways to improve a flip book so pages turn easily.

Answer: 1) Use uniform, lightweight paper stock cut to exact equal dimensions.

2) Bind the flip edge firmly using a sturdy clip or heavy staple.

QPart III: Q15: Explain two features of a correctly done mat mounting for artwork display.

Answer: 1) Clean, precise 45-degree bevel-cut inner window edges without frayed corners.

2) Balanced, uniform border margins supporting and highlighting the artwork without overlapping key details.

QPart III: Q16: Describe two actions to prevent wood breakage while carving an animal form.

Answer: 1) Always carve following the natural direction of the wood grain.

2) Use sharp gouges and gouge with controlled, gentle mallet pressure rather than forcing deep cuts.

QPart III: Q17: Visual Arts Practical Task requirements summary.

Answer: Composition depicting hills, trees/sisal plants, and a river/pathway covering 3/4 of A4 sheet. Rendered using pencil shading, blending, and smudging to demonstrate form and depth.

8. SOCIAL STUDIES - SST (GRADE 7 SBA 2026)

Q1: Explain five qualities of a global citizen.

Answer: 1) Awareness of World Affairs: Understands global interconnectedness, social justice, and environmental challenges.

2) Respect for Cultural Diversity: Values individual differences, religions, and cultures without prejudice.

3) Environmental Stewardship: Actively practices and promotes sustainable resource conservation globally.

4) Ethical Responsibility: Upholds universal human rights, peace, and social justice across borders.

5) Active Civic Participation: Engages in community and international development initiatives.

Q2: Give six points on the importance of studying Social Studies.

Answer: 1) Builds an understanding of personal history, cultural heritage, and national identity.

2) Fosters responsible citizenship, constitutional awareness, and democratic participation.

3) Enhances environmental management and spatial geography awareness.

4) Promotes peaceful conflict resolution, social cohesion, and national unity.

5) Develops critical thinking and analytical skills for solving societal challenges.

6) Opens diverse career opportunities in law, governance, environmental science, and diplomacy.

Q3: Give five factors that led to the growth of the Great Zimbabwe kingdom.

Answer: 1) Control over lucrative regional gold, copper, and ivory trade routes linking to Indian Ocean ports.

2) Fertile agricultural land and favorable climate supporting large-scale cattle pastoralism.

3) Strong centralized leadership under powerful religious and political kings.

4) Strategic geographical location providing natural defense and water access.

5) Abundant granite stone resources used to construct monumental defensive stone enclosures.

Q4: Give seven contributions of early African kingdoms to the development of the modern world.

Answer: 1) Architecture & Structural Engineering: Monumental stone building techniques (Egyptian Pyramids, Great Zimbabwe).

2) Medicine & Surgery: Early surgical procedures, mummification, and herbal pharmacology in

ancient Egypt.

3) Mathematics & Astronomy: Development of geometry, solar calendars, and counting systems.

4) Metallurgy & Metalwork: Early iron smelting (Nok culture) and advanced gold mining techniques.

5) International Trade Networks: Trans-Saharan and Maritime Indian Ocean commerce systems.

6) Higher Education & Scholarship: Establishment of world renowned academic centers like the University of Timbuktu in Mali.

7) Systems of Governance: Development of constitutional monarchies, federal kingdom structures, and community dispute resolution systems.

Q5: Give four ways in which the Constitution of Kenya promotes national development.

Answer: 1) Establishes the rule of law, order, and national governance structures.

2) Protects fundamental human rights and freedoms through the comprehensive Bill of Rights.

3) Promotes equitable resource distribution and local development through Devolution (County Governments).

4) Sets high standards for leadership integrity, transparency, and accountability.

Q6: Describe four elements of weather.

Answer: 1) Temperature: Degree of hotness or coldness of the atmosphere, measured in °C using a Six's maximum and minimum thermometer.

2) Rainfall (Precipitation): Amount of liquid water falling from clouds, measured in millimeters (mm) using a rain gauge.

3) Wind (Speed and Direction): Air in motion; direction indicated by a wind vane and speed measured in km/h using an anemometer.

4) Atmospheric Pressure: The force exerted by the weight of air on the earth's surface, measured in millibars (mb) using a barometer.

Q7: Identify four effects of rotation of the earth on human activities.

Answer: 1) Day and Night Cycle: Establishes human working schedules, resting periods, and 24-hour economic routines.

2) Standard Time Zones: Creates longitudinal time differences affecting global travel, communications, and international trade.

3) Ocean Tides: Dictates coastal fishing routines, harbor navigation, and tidal energy operations.

4) Deflection of Winds and Ocean Currents (Coriolis Effect): Shapes global climate patterns, guiding aviation flight paths and agricultural calendars.

Q8: Discuss three challenges likely to be encountered during Social Studies fieldwork.

Answer: 1) Adverse Weather Conditions: Heavy rains or extreme heat can disrupt movement and outdoor observations.

- 2) Communication & Language Barriers: Difficulty conversing with local community respondents.
- 3) Uncooperative Respondents: Refusal of community members to answer questionnaires or provide accurate historical data.